

Christian Arnel Reyes Alcantara, ECE, ECT, AAE

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RESEARCH INTERESTS

Ubiquitous computing, context-aware embedded systems, pervasive IoT, and cyber-physical systems for safety-critical and human-centered applications. Current work spans FreeRTOS-based hardware interfaces for nuclear reactor simulation and IoT-enabled smart laboratory environments.

EDUCATION

Master of Science in Electronics Engineering

Aug 2024 – present

Mapua University, Manila, Philippines · GWA: 1.18 | GPA: 3.82

Thesis: Design and Development of a TRIGA Research Reactor Simulator Based on a Legacy Control Console — integrating analog nuclear control hardware with real-time FreeRTOS firmware and UART-based simulator communication for operator training.

Bachelor of Science in Electronics Engineering

Jun 2013 – Jun 2018

Bulacan State University, Bulacan, Philippines · GWA: 1.9 | GPA: 3.09

Thesis: Eleksiklo — Web-Based Electronic Waste Management Application for Residents of Malolos City, Bulacan.

RESEARCH AND ACADEMIC EXPERIENCE

Instructor I

Jan 2024 – present

Colegio de San Juan de Letran, Manila, Philippines

- Investigating pedagogical methods that integrate AI and IoT frameworks to optimize engineering education and technical skill acquisition in laboratory environments.
- Leading development of an AI-assisted sign language recognition application for clinical healthcare settings, applying computer vision and embedded ML pipelines.
- Designing faculty development programs focused on modern R&D methodologies and advanced embedded systems design; managing departmental research directions and capstone project supervision.

Lead Electronics R&D Engineer

Aug 2022 – Dec 2023

Skyward Industries, Fiji

- Directed R&D and deployment of Industrial IoT (IIoT) sensing platforms for remote agricultural monitoring and smart building management — applying context-aware data acquisition architectures across distributed nodes, directly analogous to pervasive computing environments.
- Engineered hardware-layer optimizations for regional network infrastructures integrating fiber-optics, microwave, and copper networks for high-bandwidth cross-border communication links.

Project Technical Specialist I

Apr 2021 – Jul 2022

DOST – Advanced Science and Technology Institute, Taguig, Philippines

- Directed full-cycle development (multi-layer PCB, EMC pre-compliance, firmware) for national R&D projects; standardized technical documentation protocols for intellectual property disclosure.

- Managed lab-scale to industrial deployment transitions; led industrial liaison and authored technical disclosures for IP protection.

Project Technical Assistant III

Jul 2019 – Dec 2020

Bulacan State University, Bulacan, Philippines

- Provided technical mentorship for university-led research startups, bridging hardware prototyping and theoretical frameworks; maintained research operations during pandemic-era community quarantines through virtualized laboratory processes.

Research and Development Engineer

Oct 2018 – Jul 2019

Edge Electronics Ltd., Makati, Philippines

- Optimized photovoltaic energy-harvesting and smart-grid circuits using Altium/SIMetrix; executed experimental validation for EMC, surge, and harmonic compliance.
- Conducted failure analysis and proof-of-concept validation for power-electronic prototypes through regulatory bench-testing.

RESEARCH PUBLICATIONS

Alcantara, C. A., Abesado, A., & Panganiban, E. (2026). Development and Validation of an LQR-Optimized State-Feedback Controller for DC Motor Position Control. 12th International Conference on Electrical Engineering, Control and Robotics (EERC 2026). (To Appear in IEEE Xplore).

↳ Addresses precision control in embedded actuator systems — foundational to context-aware physical computing.

Alcantara, C. A., Reynoso, J. D., & Tose, T. B. (2026). IoT-Enabled Smart Laboratory System for Safety and Energy Optimization in Academic Environments. 10th International Conference on Robotics, Control and Automation (ICRCA 2026). (To Appear in IEEE Xplore).

↳ Demonstrates pervasive IoT deployment for real-time environmental monitoring and autonomous safety response — directly relevant to ubiquitous computing research.

CONFERENCE PRESENTATIONS

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| Apr 2026 | 12th International Conference on Electrical Engineering, Control and Robotics (EERC 2026), Xi'an Jiaotong-Liverpool University, Suzhou, China — "Development and Validation of an LQR-Optimized State-Feedback Controller for DC Motor Position Control" |
| Mar 2026 | 10th International Conference on Robotics, Control and Automation (ICRCA 2026), Osaka University Toyonaka Campus, Osaka, Japan — "IoT-Enabled Smart Laboratory System for Safety and Energy Optimization in Academic Environments" |
| Mar 2026 | International Multidisciplinary Research Conference (IMRC) — Best Presenter Award, De La Salle Araneta University, Malabon, Philippines — "IoT-Enabled Smart Laboratory System for Safety and Energy Optimization in Academic Environments" |

HONORS AND AWARDS

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| 2024 | Associate ASEAN Engineer (AAE) — Professional recognition for regional engineering excellence |
| 2021 | Global Nominee & National Finalist, NASA Space Apps Challenge |

2020	Global Finalist, Startup Elevate Contest by Founders Forum
2017	First Placer, Dalumat ng Layunin University Research Competition

TECHNICAL SKILLS & COMPETENCIES

Embedded Systems & IoT: Firmware development (C/C++, Python) for ESP32, Raspberry Pi, Arduino; FreeRTOS real-time scheduling; UART/SPI/I2C protocols; FSM design for safety-critical systems. Multi-layer PCB design (Altium, KiCAD, EasyEDA); EMC/EMI pre-compliance; high-speed circuitry; 3D modeling (SolidWorks, AutoCAD). Industrial IoT (IIoT) deployment, Building Management Systems (BMS), Digital Twin architectures, Industrial network configuration (Mikrotik, Ubiquiti).

AI, Data Science & Computer Vision: Python for machine learning, data science, and computer vision; sign language recognition pipelines; sensor fusion and data analytics for IoT systems.

Research & Scientific Communication: Research grant proposal writing, patent and prior art search, technical documentation (IEEE 830/ISO 29148), scientific paper writing and peer review.

LICENSES AND CERTIFICATIONS

- Electronics Engineer (ECE) — Professional Regulation Commission, Philippines
- Associate ASEAN Engineer (AAE) — ASEAN Federation of Engineering Organizations

REFERENCES

Available upon request.